DeepSpeed

Scalable, Memory Efficient and High Throughput Distributed Large Foundation Model Training Engine

Ma, Guokai —guokai.ma@ intel.com
Wang, Zhipeng —zhipwang@ linkedin.com

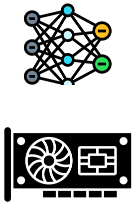
英特尔为DeepSpeed带来的助力

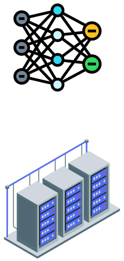
- 在DeepSpeed还是微软的研究项目的时候，英特尔与微软在DeepSpeed上有深度的合作
- 和微软合作开发了DeepSpeed抽象加速器接口（Abstract Accelerator Interface），使DeepSpeed可以轻松支持不同类型的AI加速器硬件
- 向DeepSpeed贡献了对英特尔至强处理器，英特尔独立显卡，以及Gaudi AI加速器的支持
- 是DeepSpeed自动化张量并行（AutoTP）的长期贡献者，贡献过许多流行大语言模型的AutoTP支持
- 把AutoTP引进到DeepSpeed的训练领域，使得流行的大语言模型可以直接通过HuggingFace和DeepSpeed实现张量并行训练
- 随着DeepSpeed成为PyTorch Foundation的项目之一，英特尔致力于和DeepSpeed社区成员一起，建设基于DeepSpeed的大语言模型训练生态圈

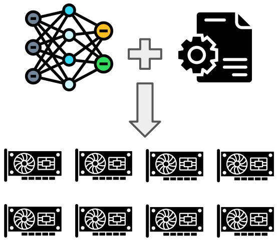
Introduction

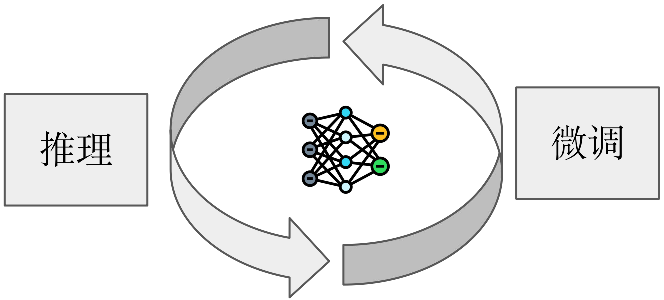
DeepSpeed的简介，使命与核心特点

- DeepSpeed是一个开源的分布式深度训练优化库，旨在为大基础模型（Large Foundation Model）提供易用，高效且易于获取的训练支持
- 我们的任务是将高性能，可伸缩，简单易用的分布式训练/微调带给各种不同的AI加速硬件
- 核心特点
  - 高性能效率: 在1到1000块以上加速卡的规模上保持高效训练
  - 易用性: 开箱即用，通过配置文件即可启用多种功能，无需修改模型代码
  - 普惠性: 在面向推理优化的硬件架构上，提供低成本的大模型微调方案

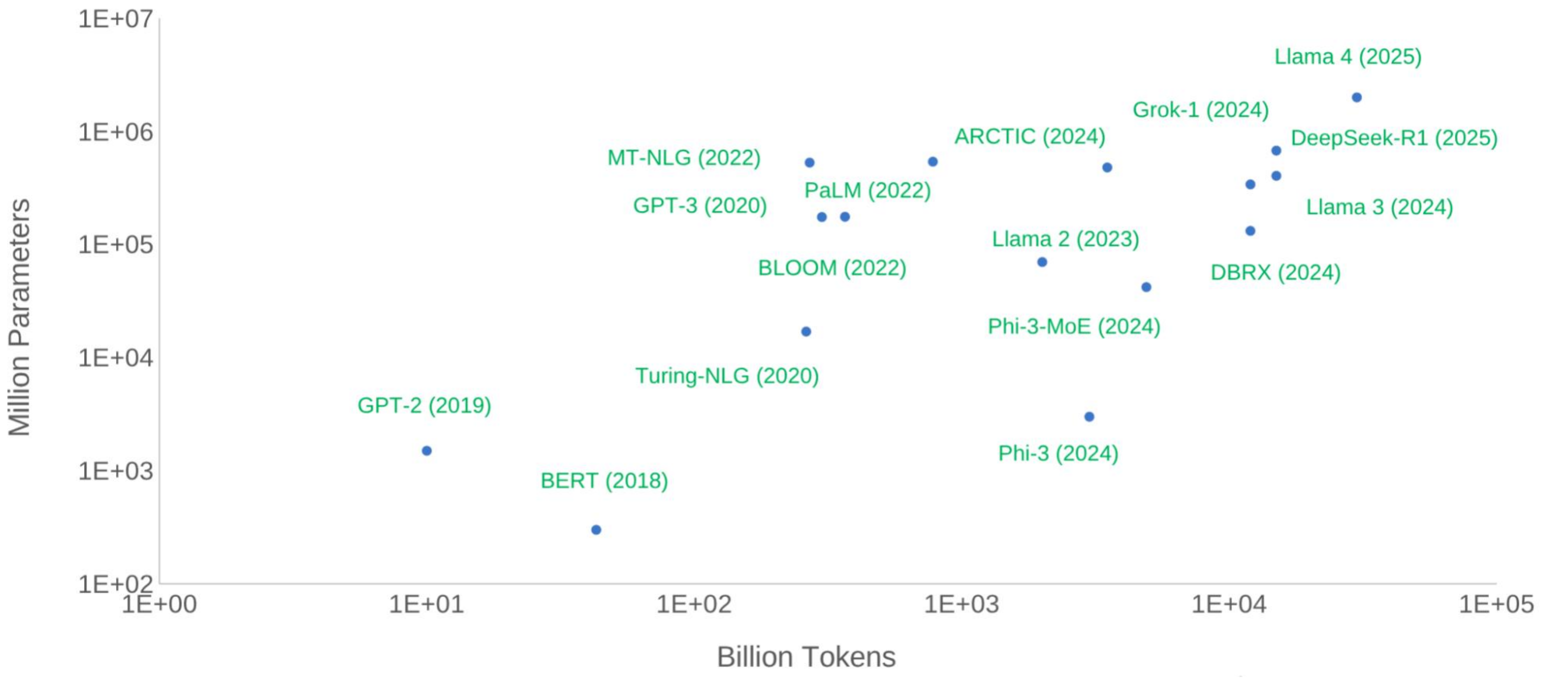




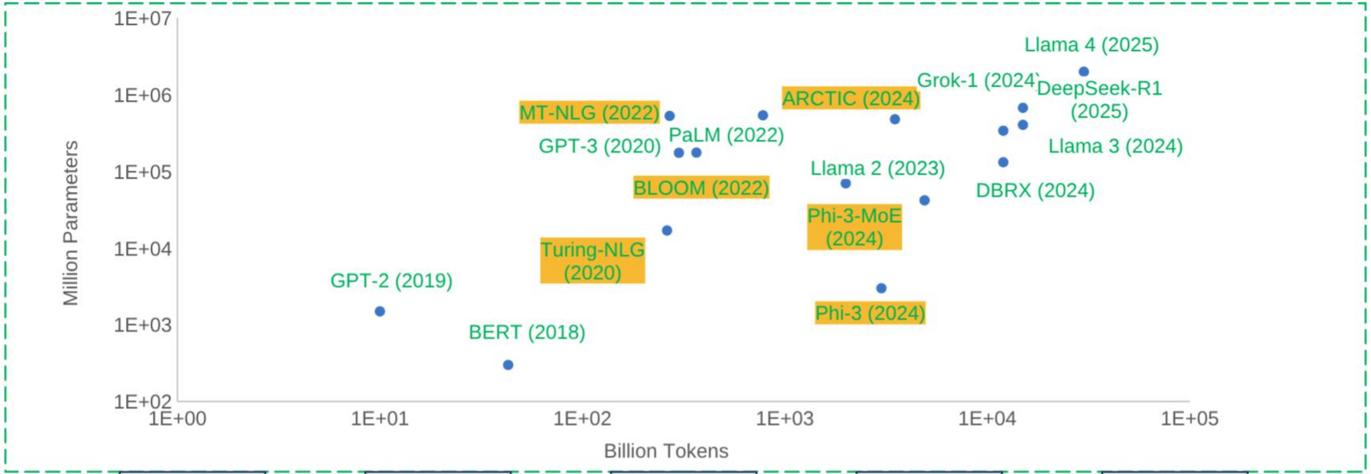


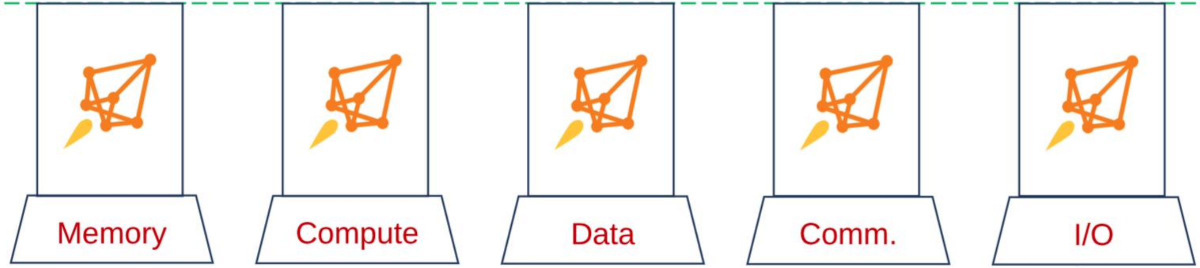


规模化驱动深度学习发展



DeepSpeed能解决深度学习规模化中的关键挑战



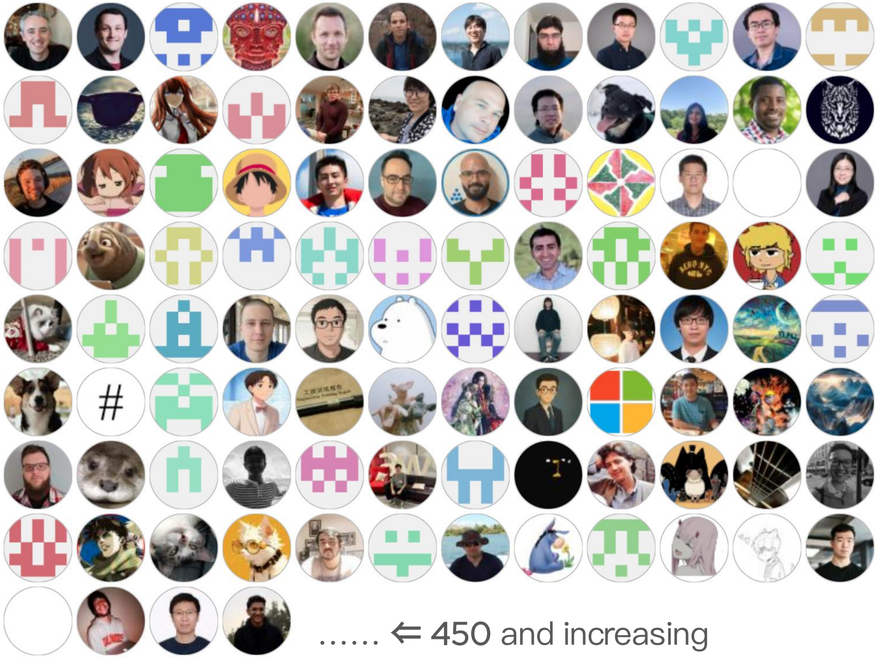


DeepSpeed支持多种类型的加速器

- DeepSpeed通过加速器抽象接口（Abstract Accelerator Interface）来提供对不同AI硬件的支持
- AI硬件可以通过实现加速器抽象接口来实现对DeepSpeed的集成
- 通过加速器抽象接口，DeepSpeed的新功能可以在不同AI硬件上很快得到支持

- 现有已经支持的加速器类型：
    - CPU
    - CUDA
    - HPU （Intel Habana Gaudi)
    - MLU （寒武纪）
    - MPS
    - NPU (华为)
    - SDAA (太初)
    - XPU （Intel独显）
    - 欢迎国内各大AI硬件厂商向DeepSpeed贡献AI硬件支持!

DeepSpeed社区贡献者



...... ⇐ 450 and increasing

DeepSpeed TSC Committers

Name	GitHub ID	Affiliation
Olatunji Ruwase	tjruwase	SnowFlake
Logan Adams	loadams	Microsoft
Masahiro Tanaka	tohtana	Anyscale
Jeff Rasley	jeffra	SnowFlake
Minjia Zhang	minjiazhang	UIUC
Ashwin Aji	ashwinma	AMD
Sam Foreman	saforem2	Argonne National Laboratory
Zhipeng Wang	PKUWZP	LinkedIn
Guokai Ma	delock	Intel

Recent features

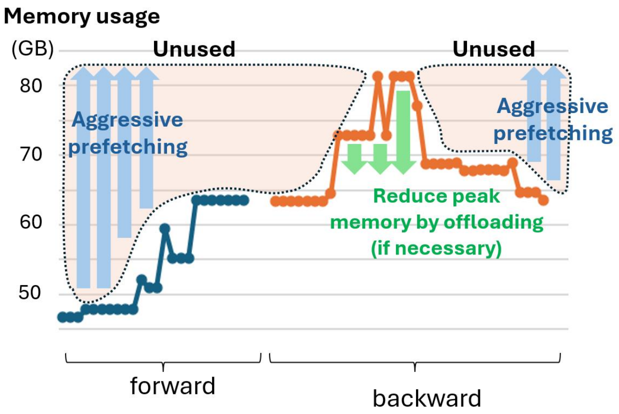
DeepSpeed vs. FSDP

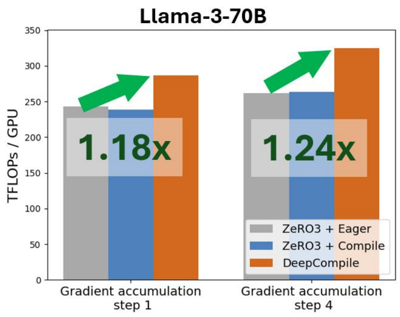
DeepSpeed offers several advantages over FSDP:

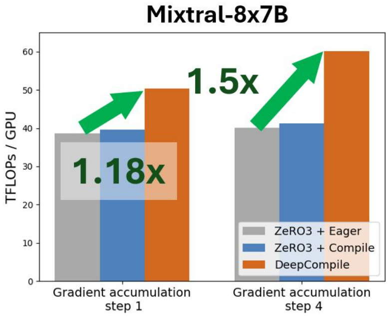
- **Industry-leading CPU/GPU offloading techniques**, much faster than FSDP2 offload, enables very large model on GH200 with high MFU (Model Flops Utilization)
- **AutoTP**, much higher performance for some dense models
- **Industry-leading sequence parallelism** Techniques for long-context model training
- **Universal checkpointing** that allows flexible number of GPUs

DeepCompile: Unlocking Compiler Optimization for Distributed Training

- DeepCompile is optimization through compiler transformations
  - Automatic Parallelization
  - Profile-guided performance tuning
  - Build on torch.compile
- Current optimizations
  - ZeRO3 + automatic communication scheduling
  - Advanced Offloading

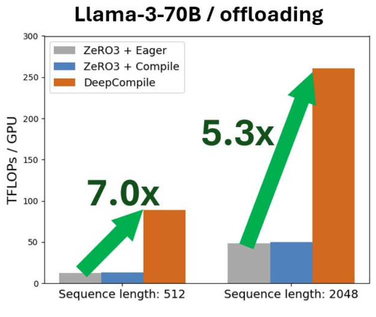






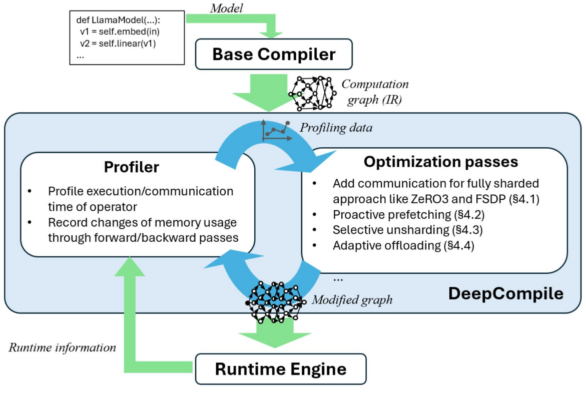
*Batch size/GPU: 1, sequence length: 1024

32 GPUs



Batch size/GPU: 1, sequence length: 512, 2048

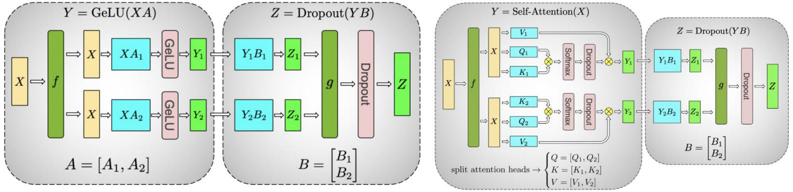
16 GPUs



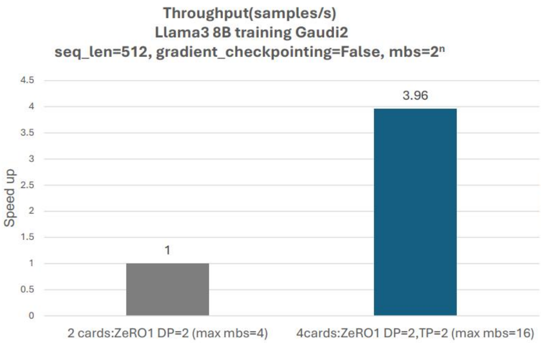
https://arxiv.org/abs/2504.09983

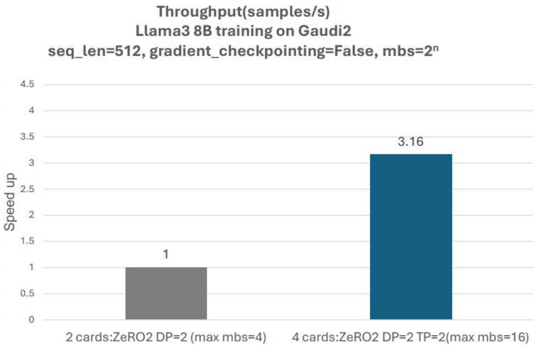
AutoTP training for HuggingFace models

- AutoTP -- Tensor parallel without model and weight changes
- Automatic Tensor Parallel for HuggingFace models
  - Analyze model config of a HF model and shard model attention heads, MLP, etc. among accelerators
  - Auto inject communication into model computation
  - No modeling change, no preprocessing of model checkpoint required
- DeepSpeed now brings AutoTP to training
  - Turn on with simple configuration change
  - Training checkpoint directly usable
  - Free up more memory for batch scaling



(a) MLP (b) Self-Attention



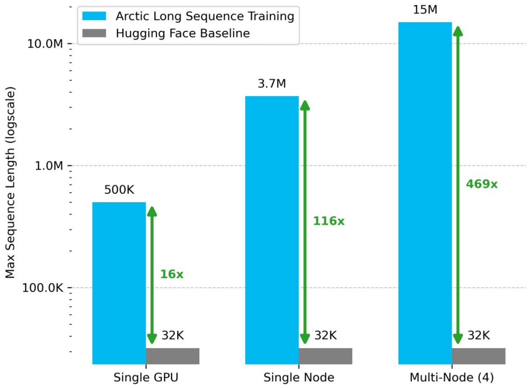


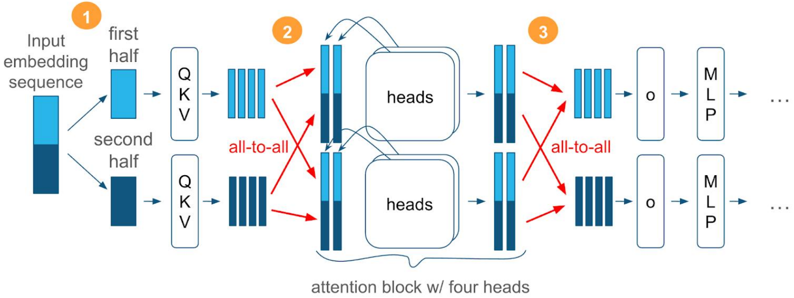

```
"ZeRO_optimization": {
  "stage": 1,
  "gather_16bit_weights_on_model_save": true,
  ...
},
"tensor_parallel":{
  "autotp_size": 4
},
```

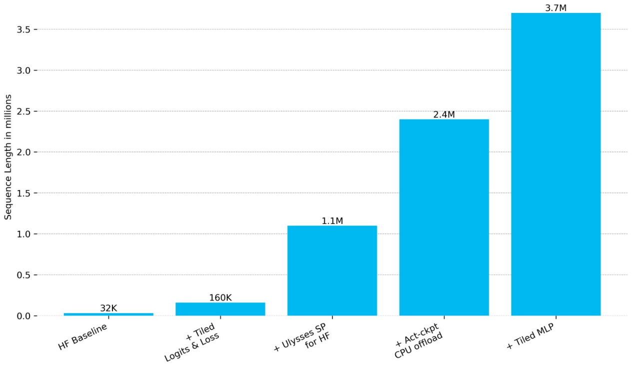

https://github.com/deepspeedai/DeepSpeed/blob/master/blogs/huggingface-tp/README.md

Arctic Long Sequence Training (ALST) - Scalable and efficient training for multi-million token sequences

- AI applications like RAG, Agentic AI, multi-modality support needs long sequence capability
- Activation memory is the primary bottleneck of long sequence training
- ALST enables long sequence on small scale infrastructure with the following technologies:
  - Ulysses SP
  - Sequence titling on Logits+loss and MLP
  - Checkpoint offload







https://www.arxiv.org/abs/2506.13996

Muon optimizer

- **Muon** (**M**oment**U**m **O**rthogonalized by **N**ewton-Schulz) is an optimizer for 2D parameters of neural network hidden layers
- **~1.33x training speed up** compares to adam optimizer when training 1.5B transformer
- Muon is the optimizer used by Kimi on **Moonlight-16B-A3B**
- DeepSpeed integrated Muon optimizer to work with **ZeRO stage 1&2** through **tensor unflattening**
- In DeepSpeed, you can change optimizer type to Muon by a line of change in config file
- Future work
  - Support Muon in **CPU offload**
  - Support **ZeRO stage 3**
  - More performance efficient implementations

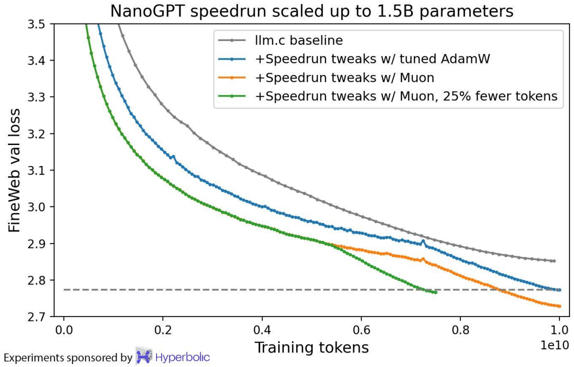
https://kellerjordan.github.io/posts/muon/

```
"optimizer": {
  "type": "Muon",
  "params": {
    "lr": 2e-5,
    "betas": [0.9, 0.999],
    "eps": 1e-8,
    "weight_decay": 0.01
  }
},
```

Algorithm 2 Muon

Require: Learning rate η, momentum μ

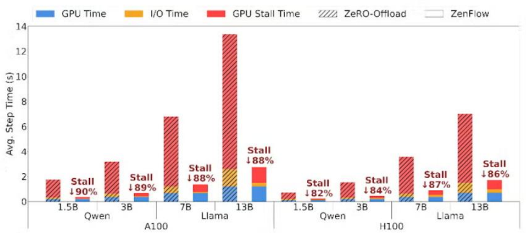
1: Initialize $B_0 \leftarrow 0$
2: **for** $t = 1, \ldots$ **do**
3: Compute gradient $G_t \leftarrow \nabla_\theta \mathcal{L}_t(\theta_{t-1})$
4: $B_t \leftarrow \mu B_{t-1} + G_t$
5: $O_t \leftarrow \mathrm{NewtonSchulz5}(B_t)$
6: Update parameters $\theta_t \leftarrow \theta_{t-1} - \eta O_t$
7: **end for**
8: **return** θ_t

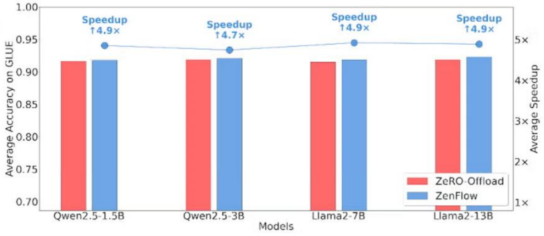


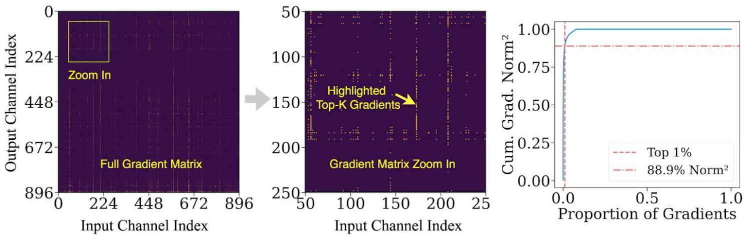
ZenFlow: Stall-Free Offloading Engine for LLM Training

- Offloading is a widely used technique to mitigate GPU memory pressure caused by large LLM sizes
- Traditional offloading frameworks suffer from GPU stalls due to offloading computation on slower CPUs
- ZenFlow prioritize important gradients for GPU updates and defer the rest for async CPU-side accumulation
  - A few important gradients are updated on GPU
  - Most accumulated non-important gradients will be updated less frequently on CPU
  - CPU updates weights in parallel to GPU computation, makes CPU overhead hidden in GPU time.
- Up to **5x** end-to-end speed-up over ZeRO-Offload and **6.3x** over ZeRO-Infinity without sacrifice accuracy

```
"zenflow": {
    "topk_ratio": 0.1,
    "update_interval": 4,
    "full_warm_up_rounds": 0,
    "overlap_step": true
}
```







Offloading with No Compromise	Unified Support for Single & Multi-cards	Seamless Intergration with Deepspeed	Near Zero-Stall GPU Execution

Top 1% of gradients may contribute over 85% of gradient norms

https://arxiv.org/abs/2505.12242

Future plan

技术路线

长期目标 (North Star): 将 DeepSpeed 打造为业界广泛默认的加速器上的Large Foundation Model训练引擎，在各硬件平台上均实现顶尖性能。

下一季度计划开展多项新技术：

- **扩展加速器支持：**
  实现对 XLA/TPU 及更多类型加速器（如 AWS Trainium）的 DeepSpeed 支持。
- **强化学习与大模型训练整合：**
  当前大语言模型（LLM）训练后阶段的强化学习（RL）在业界广泛应用。DeepSpeed 将与字节跳动团队开发的开源强化学习库 verl 深度合作，将 DeepSpeed 集成为其训练后端。verl 用户将可借此利用 DeepSpeed 所提供的更高效的 CPU offloading、AutoTP、序列并行和通用检查点功能。
- **支持新兴模型架构训练加速：**
  扩展 DeepSpeed 能力至扩散语言模型、多模态基础模型等新型架构的训练加速。
- **混合专家模型（MoE）训练加速：**
  持续优化并推进 MoE 模型的高效分布式训练。

社区构建与生态拓展

DeepSpeed 现已加入 PyTorch 基金会!

- 我们希望构建全球化的社区，并诚挚欢迎来自人工智能建模、机器学习系统、应用等领域的 researchers、工程师与实践者加入我们的社区，一起探索与共创!
- 加入我们的 **Slack 频道**（链接详见 DeepSpeed 官网），欢迎随时在官网提交 issue 和 PR。
- 我们将定期组织线上见面会和开发者会议，分享我们的技术路线图与开发计划。
- 我们也将通过博客文章与学术论文，分享 DeepSpeed 项目的研究与开发成果。诚邀社区中的每一位伙伴参与其中!

国内社区建设

- 中国是AI发展的核心引擎，我们期待在中国建立壮大一个充满活力的DeepSpeed开发者社区!
- 我们将开通微信公众号及微博账号，并在知乎平台发布更多中文技术博客。
- 定期举办线下DeepSpeed技术沙龙与开发者交流会
- 更广泛地与中国本土AI社区及企业开展深度合作，共同推动技术进步与生态繁荣

Thanks!

Backup

技术路线

North Star: *Making DeepSpeed the default choice of Foundation Models Training Engine in the industry on a wide range of accelerators with top-notch performance.*

Several new technical initiatives are planned out in the upcoming quarter:

1. DeepSpeed support on XLA/TPU and other types of accelerators (e.g. AWS Trainium).
2. Reinforcement Learning for LLM post training is widely practiced in the industry, ***DeepSpeed will work closely with verl, the popular OSS RL library developed by the Bytedance team, to integrate DeepSpeed as the training backend***. This can enable verl users to benefit from faster CPU/GPU offloading, AutoTP, Sequence Parallelism and Universal Checkpointing features developed by the DeepSpeed team.
3. Extending DeepSpeed's capabilities to speedup training of novel model architectures (e.g. Diffusion Language Models, Multimodal Foundation Models etc.).
4. Mixture-of-Expert (MoE) model training acceleration

Community Building & Outreach

DeepSpeed now joins the PyTorch Foundation!

- We want to build communities across the world, we welcome researchers, engineers and practitioners across AI modeling, ML Systems, Applications etc. to join our community to hack together!
- Join our slack channel (link is on DeepSpeed homepage), and feel free to raise issues/PRs.
- We will organize regular meetups and developer meetings to share our technical roadmaps and development plan.
- We will publish blog posts and academic articles to present the research and development work form DeepSpeed project. Welcome any participation from the community!

Community Building in China

- China is the powerhouse of AI development, we would like to grow and foster a strong developer community for DeepSpeed in China!
- We will set up the Weixin public account and Weibo account, and publish more technical blog posts in Chinese on Zhihu.
- We will organize local DeepSpeed meetups and developer meetings
- We will work more extensively with local AI communities and companies in China to facilitate deeper collaboration and technical development.

Thanks